Wonder of You — Technical Design

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Overview

Wonder of You is a six-act Unity VR experience. It runs standalone on Meta Quest 2, Quest 3 and Quest 3S at 72 Hz, and a full playthrough takes 15–25 minutes. Tina Jiang built it alone: design, scripting, scene building and audio.

The technical goal was a quiet one. The player should never notice the systems. Doors open only when the story is ready, voices come from the right place in the room, and every scene change feels like a slip in memory, not a loading screen.

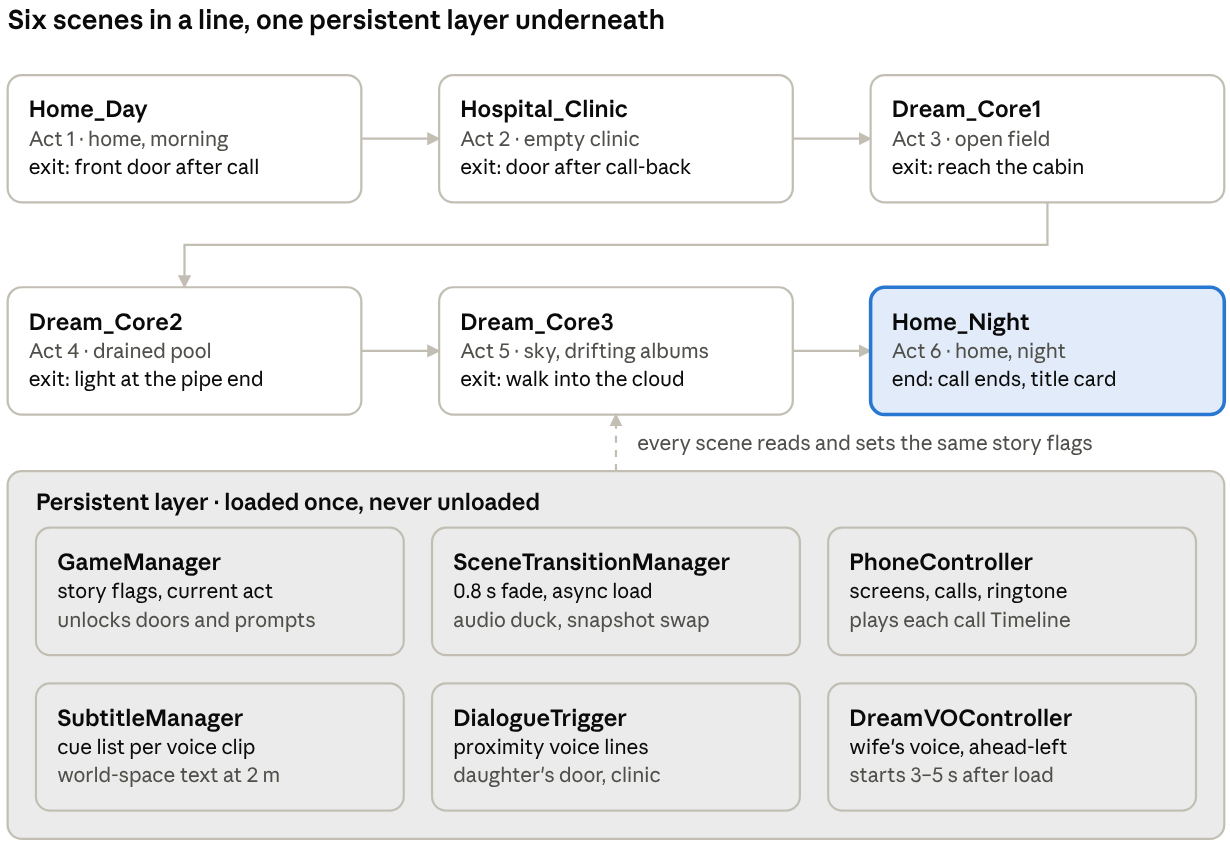
Engine	Unity 6000.3.13f1, Universal Render Pipeline
XR stack	OpenXR with the Meta OpenXR feature group, XR Interaction Toolkit
Target devices	Meta Quest 2, Quest 3, Quest 3S (Android, ARM64, IL2CPP)
Frame rate target	72 Hz on every device. Quest 2 sets the performance budget.
Input	Controllers: grab, poke on the phone screen, left-stick continuous movement
Sequencing	Timeline for cutscene beats, C# state flags for story gates
Audio	Unity AudioMixer, 3D spatialised voice, snapshot transitions
Text	TextMeshPro on world-space canvases, English only

Supporting three headsets. The project uses OpenXR and the XR Interaction Toolkit and never calls a device-specific API, so a single Android build runs on all three headsets. Quest 2 has the weakest GPU (Snapdragon XR2 Gen 1), so it sets the polygon, texture and

lighting budgets. Quest 3 and 3S (XR2 Gen 2) run the same content with headroom to spare.

Architecture

The experience is six Unity scenes, one per act, played in a fixed order. A persistent layer of managers lives in a bootstrap scene and stays loaded the whole time. Only one act scene is in memory at once, which keeps memory use low on Quest 2.



scene flow and persistent managers · 6 scenes, 6 systems

Each act scene holds only its environment, props and triggers. Story logic lives in the persistent layer, so an act scene can be rebuilt or relit without breaking progress.

Story state. `GameManager` is a singleton that keeps a small set of boolean flags, for example `clockPickedUp`, `morningCallDone`, `missedCallsSeen` and `callBackDone`. Interactables raise UnityEvents. The manager records each flag and broadcasts a C# event, which doors, prompts and the phone listen for. Gates are written as flag checks, not timers, so the story can never move on before the player has heard what they need to.

Interactables. All props share an `InteractableObject` base built on XR Interaction Toolkit components. Designers wire each prop's grab, release or trigger to a flag in the Inspector, without writing a script for each prop.

Interaction

Every interaction is either something Dad would do on an ordinary day, or simply walking. There are no menus, inventories or puzzles.

Interaction	Where	How it works
Pick up an object	Clock, pill bottle, patient file (Acts 1–2)	`XRGrabInteractable` with physics. On grab it fires a UnityEvent that sets a story flag.
Answer or return a call	Phone (Acts 1, 2, 6)	The phone is a grab prop with a world-space canvas on its screen. Buttons respond to a poke interactor. `PhoneController` changes screens: lock, incoming, missed calls with message, calling, in call.
Overhear	Daughter's door (Act 1)	A trigger collider plays the door dialogue when Dad comes close. No button press.
Look	Calendar, patient board (Act 2)	Nothing is required. These details are there for players who stop and read.
Open a door	Front door (Act 1), clinic exit (Act 2)	The door stays locked until its story flag is set, so the player can't leave before a call has finished.
Walk toward a light	Dream scenes (Acts 3–5)	A large invisible sphere trigger, 1.5–2 m in radius, sits around each landmark and starts the scene change.

Prompts. Dad's inner reminders appear as soft white text on world-space canvases next to the object they refer to. For example, "The clock on the nightstand" appears with a small arrow beside the clock. Each prompt hides itself once its action is done. There is no screen-space UI, and the dream scenes have no prompts at all.

Subtitles. Each voice clip has a ScriptableObject cue list of start time, end time and line. `SubtitleManager` shows the current line on a TextMeshPro panel 2 m ahead of the camera and 0.3 m below eye level: white text on a translucent black plate. It hides after 0.5 s of silence. Dad's mumbled inner lines are left unsubtitled on purpose.

Movement and comfort. The player moves with continuous left-stick movement at 0.8 m/s and turns with snap turn. Teleport is disabled, because each scene asks the player to walk toward something. A comfort vignette is on by default and can be turned off. In the dream scenes the walkable path narrows toward the landmark, which guides the player forward without any visible wall.

Audio, transitions and Timeline

Sound carries most of the story, so the mixer is set up around the question of who is speaking and how real they are.

Voice or layer	Treatment	Why
Wife on the phone (Acts 1–2)	3D source on the phone. A faint high-pass filter and reverb on a mixer group.	She sounds almost real. A careful listener hears that something is off.
Wife in the dreams (Acts 3–5)	No filter. A 3D source moved by `DreamVOController` to sit about 1 m ahead and to the left of the camera. It starts 3–5 s after the scene loads.	It sounds as if she is walking beside him.
Daughter, adult (Act 6)	Clean 3D source on the phone.	The one voice in the piece that is fully real.
Daughter, child (Act 1)	3D source behind her bedroom door, slightly muffled.	A memory heard through a door.
Dad's inner voice	2D, no spatialisation, quiet.	It plays inside the player's head.
Ringtone, *The Wonder of You*	3D source on the phone. It loops until answered. Act 6 uses a louder mix.	The motif that links the three calls.

Voice or layer	Treatment	Why
Ambience	One loop per scene: birdsong and fridge hum, fluorescent buzz, wind, dripping water, crickets and a ticking clock.	Each place is recognisable with your eyes closed.

Each act has an AudioMixer snapshot, and `TransitionTo` blends between them over the scene fade. In the clinic, the footsteps follow the player's movement speed, so they stop when the player stops.

Scene transitions. Every act change uses the same sequence:

1. The player reaches a trigger: a door, or the sphere around a dream landmark.
2. `SceneTransitionManager` fades a black canvas in front of the camera from alpha 0 to 1 over 0.8 s, and ducks the audio.
3. Halfway through the fade it calls `LoadSceneAsync` for the next act and unloads the previous one.
4. It fades back from 1 to 0 over 0.8 s, and `GameManager` sets the new act's flags.

The loading is hidden inside the darkness, so a scene change feels like losing time.

Timeline. Each call (incoming ring, answer, voice, hang-up, next prompt) is one Timeline asset with signal tracks. The signals tell `PhoneController`, `SubtitleManager` and the prompts when to act. The ending is also a Timeline: the call ends, 3 s of black, then the title card fades in over 2 s, holds for 5 s and fades out. Timing can be adjusted in the editor without changing code.

Performance and testing

All of the budgets are set for Quest 2. If a scene holds 72 Hz there, it holds on Quest 3 and 3S too.

Rendering

- URP with 4x MSAA. HDR and post-processing are off, and ambient occlusion is baked.
- Lighting is baked in every scene. Each scene has at most one real-time light: the flickering clinic tube, or the warm lamp at night.
- Textures use ASTC compression. Static geometry is marked Static so it can batch, and the multi-room house uses baked occlusion culling.

- Larger props have LODs. The dream spaces depend on fog and gradient skies more than geometry, which keeps them cheap even though they feel endless.
- Transparent dream materials, such as the clouds and drifting albums, use an explicit render queue so they don't sort badly against opaque geometry.

Build

- One Android build: ARM64, IL2CPP, minimum API level 29, linear colour space.
- Development builds are installed over wireless ADB through Meta Quest Developer Hub. Layout and logic are checked first in the Meta XR Simulator and the XR Device Simulator in the editor.

Test checklist

- [ ] Holds 72 Hz on Quest 2, Quest 3 and Quest 3S in every act (Unity Profiler and OVR Metrics Tool)
- [ ] Full playthrough on each headset is 15–25 minutes, with no soft-locks
- [ ] Front door opens only after the morning call. Clinic exit opens only after the call-back.
- [ ] Skipping the pill bottle never blocks progress
- [ ] The ringtone plays at all three cues, and the Act 6 mix is the loudest
- [ ] The wife's phone voice is filtered. Her dream voice is dry and sits ahead and to the left.
- [ ] Subtitles match the recorded lines and are readable at 2 m
- [ ] Every fade hides its load, with no flash or hitch
- [ ] The comfort vignette can be turned on and off
- [ ] All on-screen and in-world text is English: prompts, phone, calendar, file, label, end card
- [ ] The end card fades in, holds and fades out cleanly